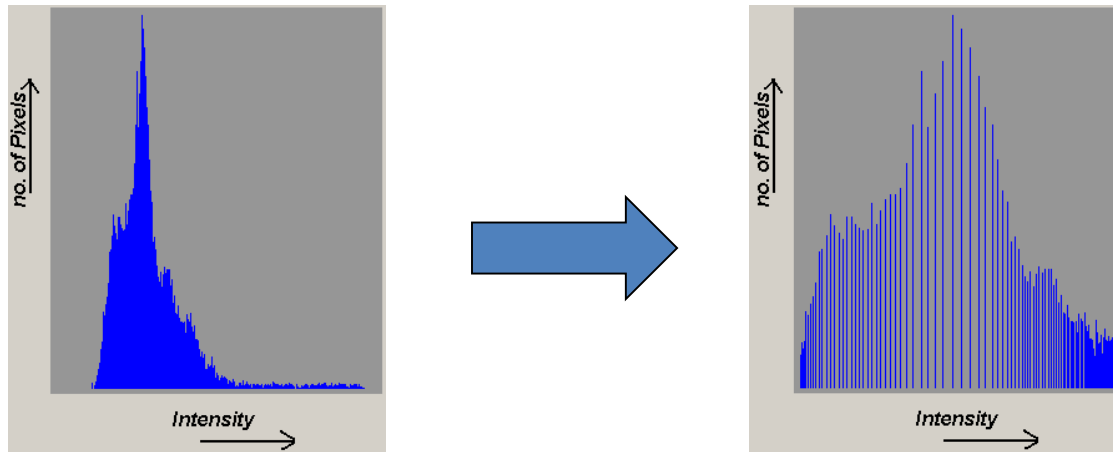


Digital Image Processing

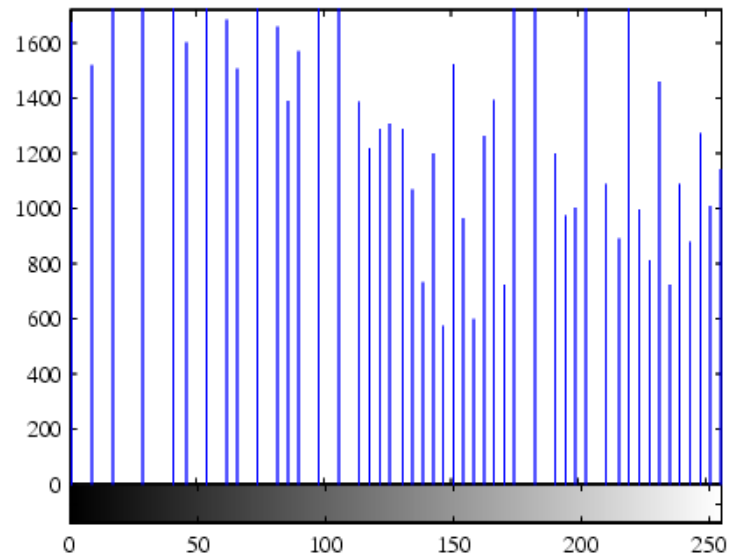
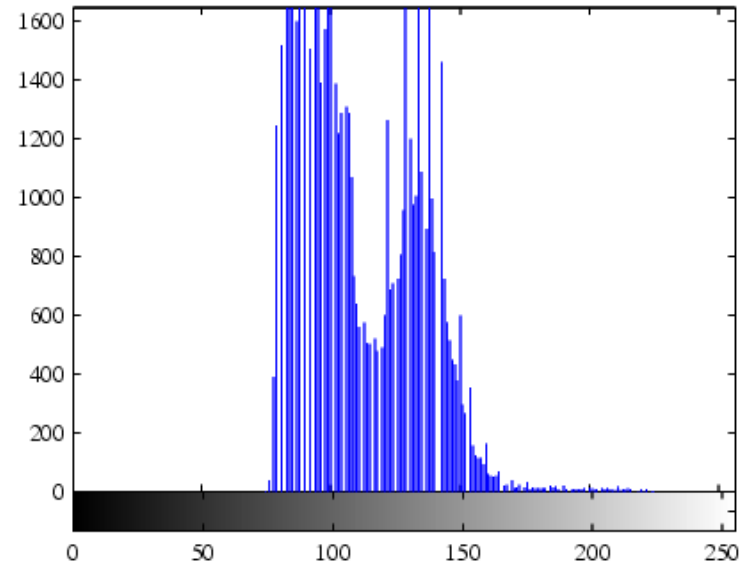
Lecture # 4 **Histogram Equalization & Matching**

Histogram Equalization

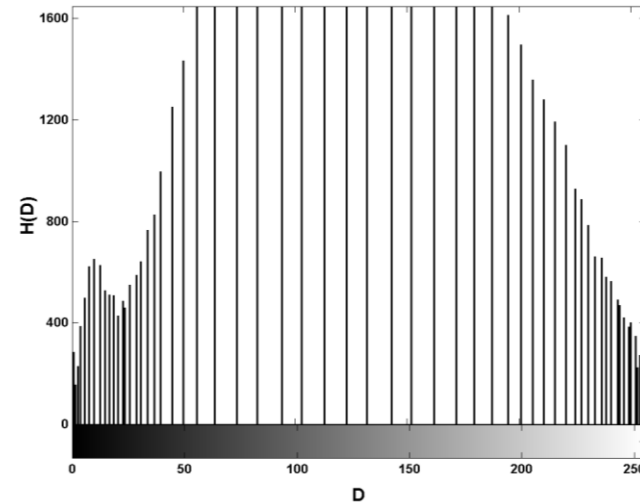
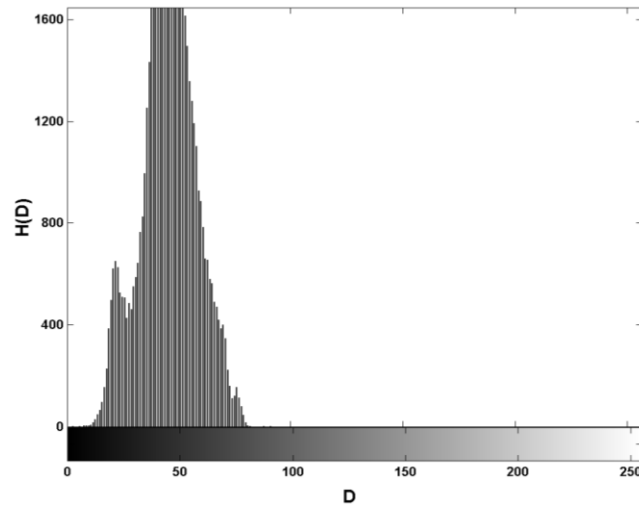
Histogram equalization re-assigns the intensity values of pixels in the input image such that the output image contains a uniform distribution of intensities



HISTOGRAM EQUALIZATION



AERIAL PHOTOGRAPH OF THE PENTAGON



**Resulting image uses more of dynamic range.
Resulting histogram almost, but not completely, flat.**

The Probability Distribution Function of an Image

$$\text{Let } A = \sum_{g=0}^{255} h_I(g)$$

Note that since $h_I(g)$ is the number of pixels in I with value g ,

A is the number of pixels in I . That is if I is R rows by C columns then $A = R \times C$.

Then,

$$p_I(g) = \frac{1}{A} h_I(g)$$

This is the probability that an arbitrary pixel from I has value g .

The Probability Distribution Function of an Image

- $p(g)$ is the fraction of pixels in an image that have intensity value g .
- $p(g)$ is the probability that a pixel randomly selected from the given image has intensity value g .
- Whereas the sum of the histogram $h(g)$ over all g from 0 to 255 is equal to the number of pixels in the image, the sum of $p(g)$ over all g is 1.
- p is the **normalized histogram** of the image

The Cumulative Distribution Function of an Image

Let $\mathbf{q} = I(r,c)$ be the value of a randomly selected pixel from I . Let g be a specific gray level. The probability that $q \leq g$ is given by

$$P_I(g) = \sum_{\gamma=0}^g p_I(\gamma) = \frac{1}{A} \sum_{\gamma=0}^g h_I(\gamma) = \frac{\sum_{\gamma=0}^g h_I(\gamma)}{\sum_{\gamma=0}^{255} h_I(\gamma)},$$

where $h_I(\gamma)$ is the histogram of image I .

This is the probability that any given pixel from I has value less than or equal to g .

The Cumulative Distribution Function of an Image

Let $\mathbf{q} = I(r,c)$ be the value of a randomly selected pixel from I . Let g be a specific gray level. The probability that $q \leq g$ is given by

$$P_I(g) = \sum_{\gamma=0}^g p_I(\gamma) = \frac{1}{A} \sum_{\gamma=0}^g h_I(\gamma) = \frac{\sum_{\gamma=0}^g h_I(\gamma)}{\sum_{\gamma=0}^{255} h_I(\gamma)},$$

where $h_I(\gamma)$ is the histogram of image I .

Also called CDF for "Cumulative Distribution Function".

This is the probability that any given pixel from I has value less than or equal to g .

The Cumulative Distribution Function of an Image

- $P(g)$ is the fraction of pixels in an image that have intensity values less than or equal to g .
- $P(g)$ is the probability that a pixel randomly selected from the given band has an intensity value less than or equal to g .
- $P(g)$ is the cumulative (or running) sum of $p(g)$ from 0 through g inclusive.
- $P(0) = p(0)$ and $P(255) = 1$;

Histogram Equalization

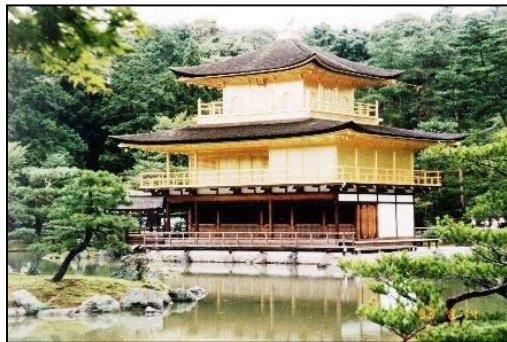
Task: remap image I so that its histogram is as close to constant as possible

Let $P_I(\gamma)$

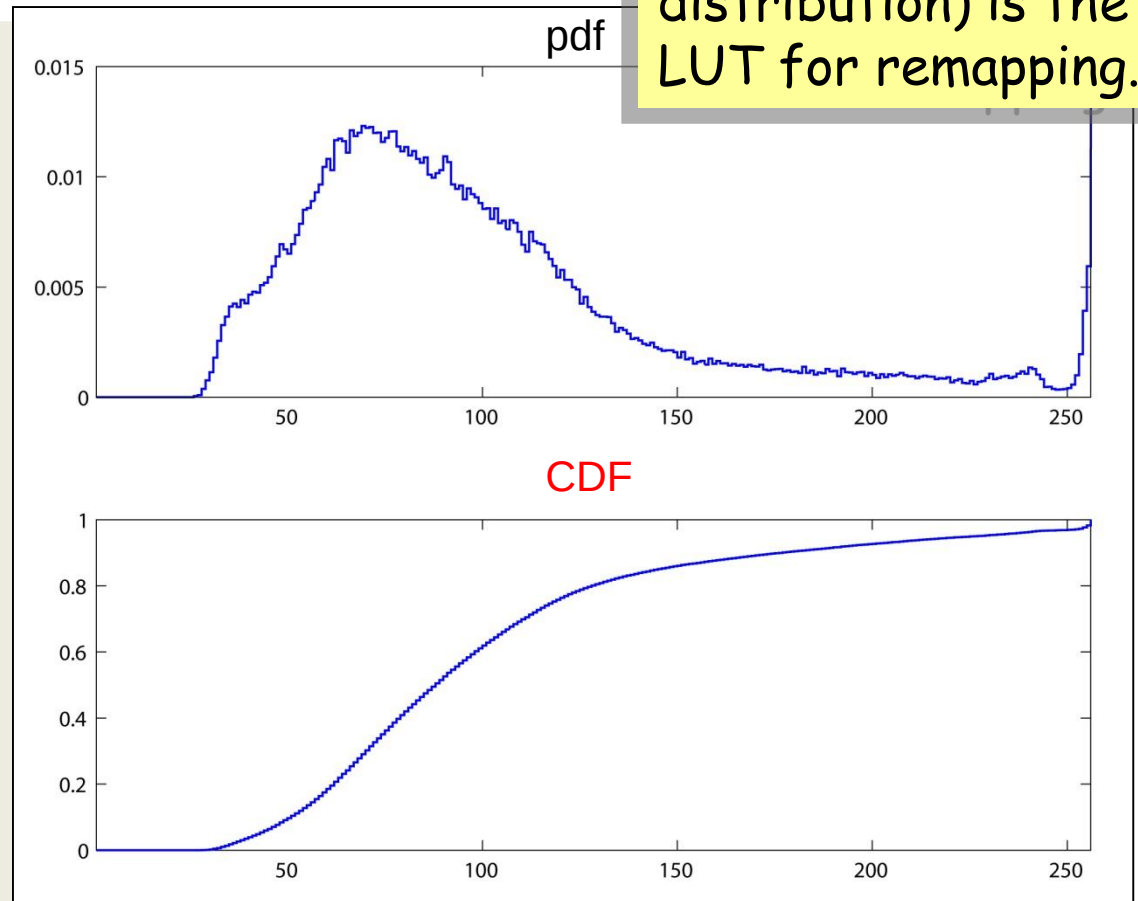
be the cumulative (probability) distribution function of I .

The CDF itself is used as the LUT.

Histogram Equalization



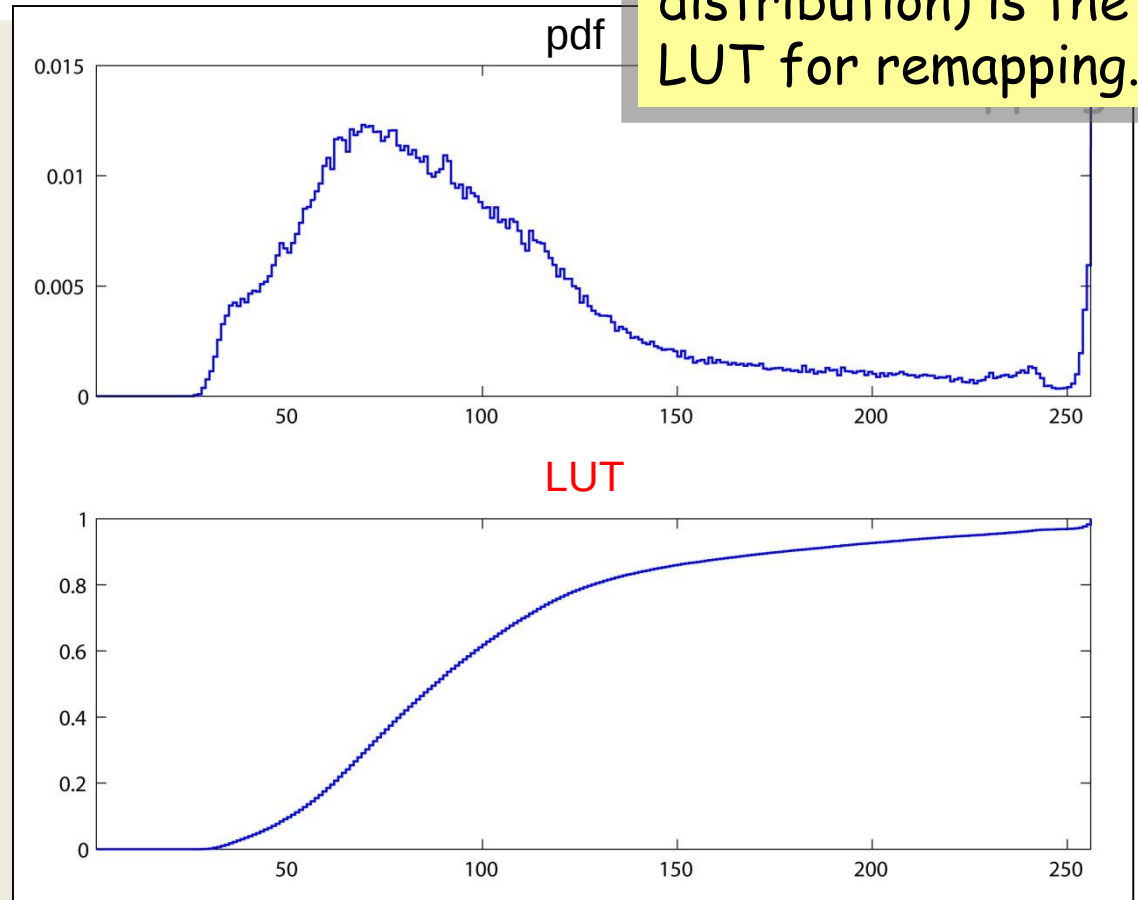
The CDF (cumulative distribution) is the LUT for remapping.



Histogram Equalization



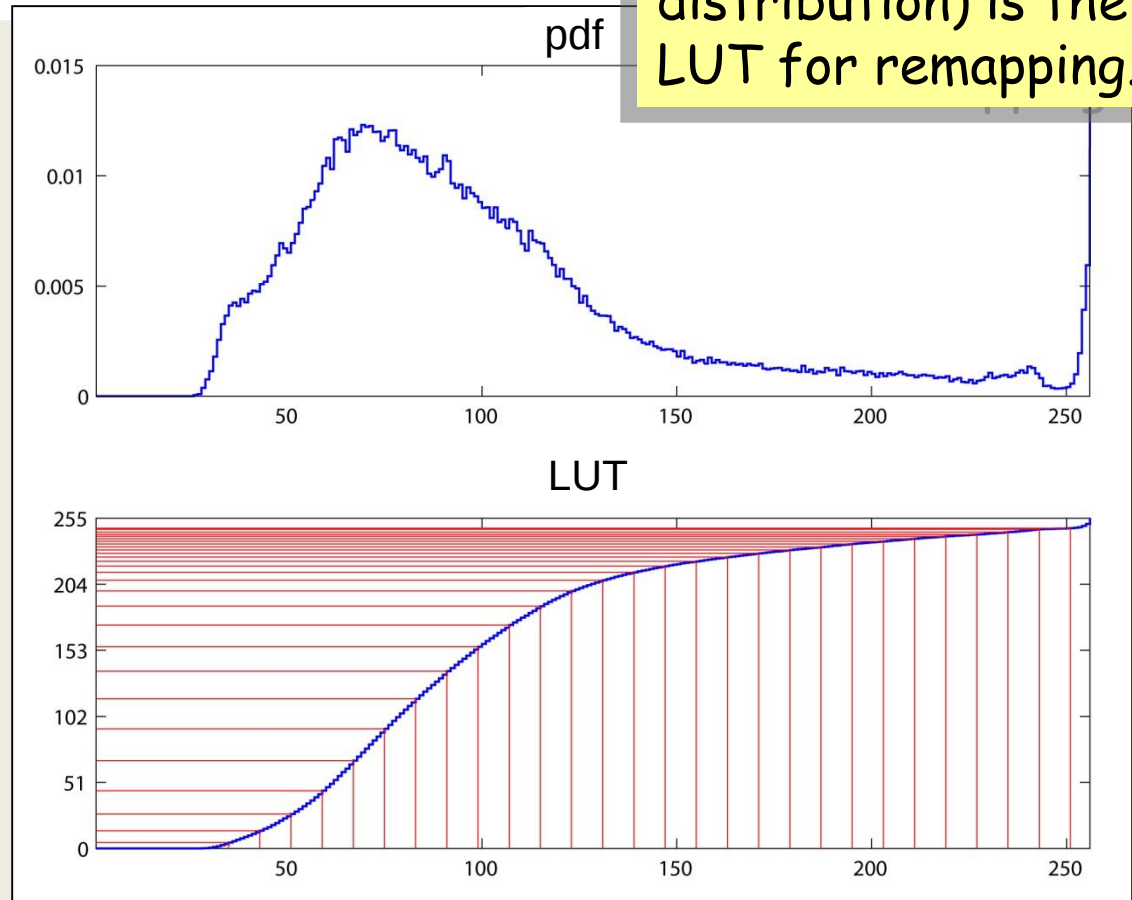
The CDF (cumulative distribution) is the LUT for remapping.



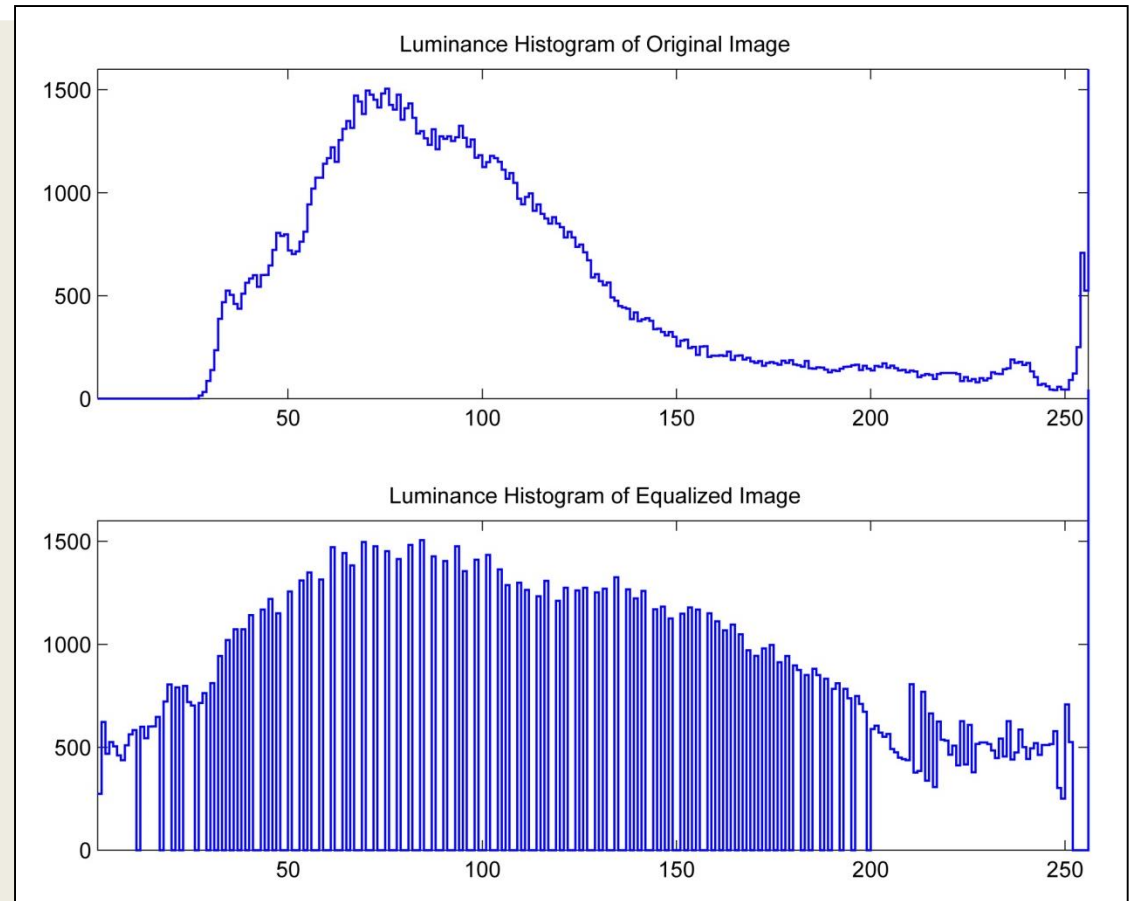
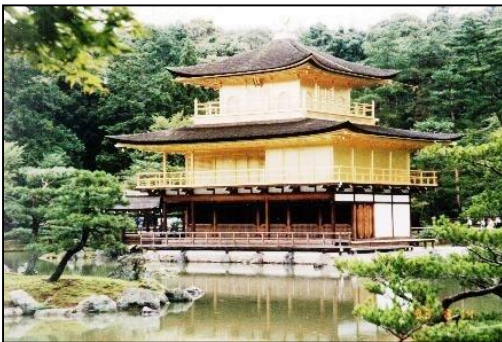
Histogram Equalization



The CDF (cumulative distribution) is the LUT for remapping.



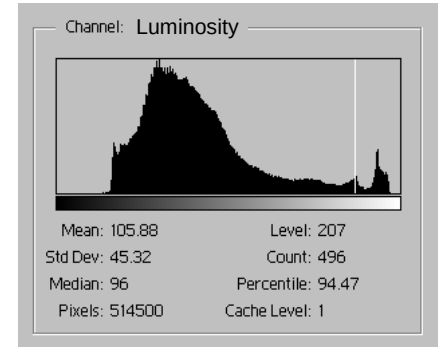
Histogram Equalization



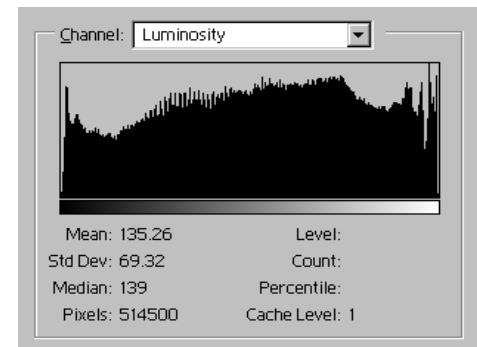
Histogram Equalization



$$J(r,c) = 255 \cdot P_I[I(r,c)].$$



before



after

HISTOGRAM EQUALIZATION IMPLEMENTATION

0	0	0	0	0
1	1	1	1	4
4	5	6	6	6
8	8	8	8	9

2	2	2	2	2
4	4	4	4	5
5	5	7	7	7
9	9	9	9	9

Gray levels	0	1	2	3	4	5	6	7	8	9
Counts ($h(r_k)$)	5	4	0	0	2	1	3	0	4	1
	r_0	r_1			r_2	r_3	r_4		r_5	r_6
Normalized h ($P(r_k)$)	5/20	4/20	0	0	2/20	1/20	3/20	0	4/20	1/20
cdf $F(r_k)$	5/20	9/20			11/20	12/20	15/20		19/20	20/20
$s_k = \text{round}(9 \cdot F(r_k))$	2	4			5	5	7		9	9
	s_0	s_1			s_2	s_3	s_4		s_5	s_6

Histogram Equalization: Example



52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

An 8x8 image



Histogram Equalization: Example

Fill in the following table/histogram

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52		64		72		85		113	
55		65		73		87		122	
58		66		75		88		126	
59		67		76		90		144	
60		68		77		94		154	
61		69		78		104			
62		70		79		106			
63		71		83		109			

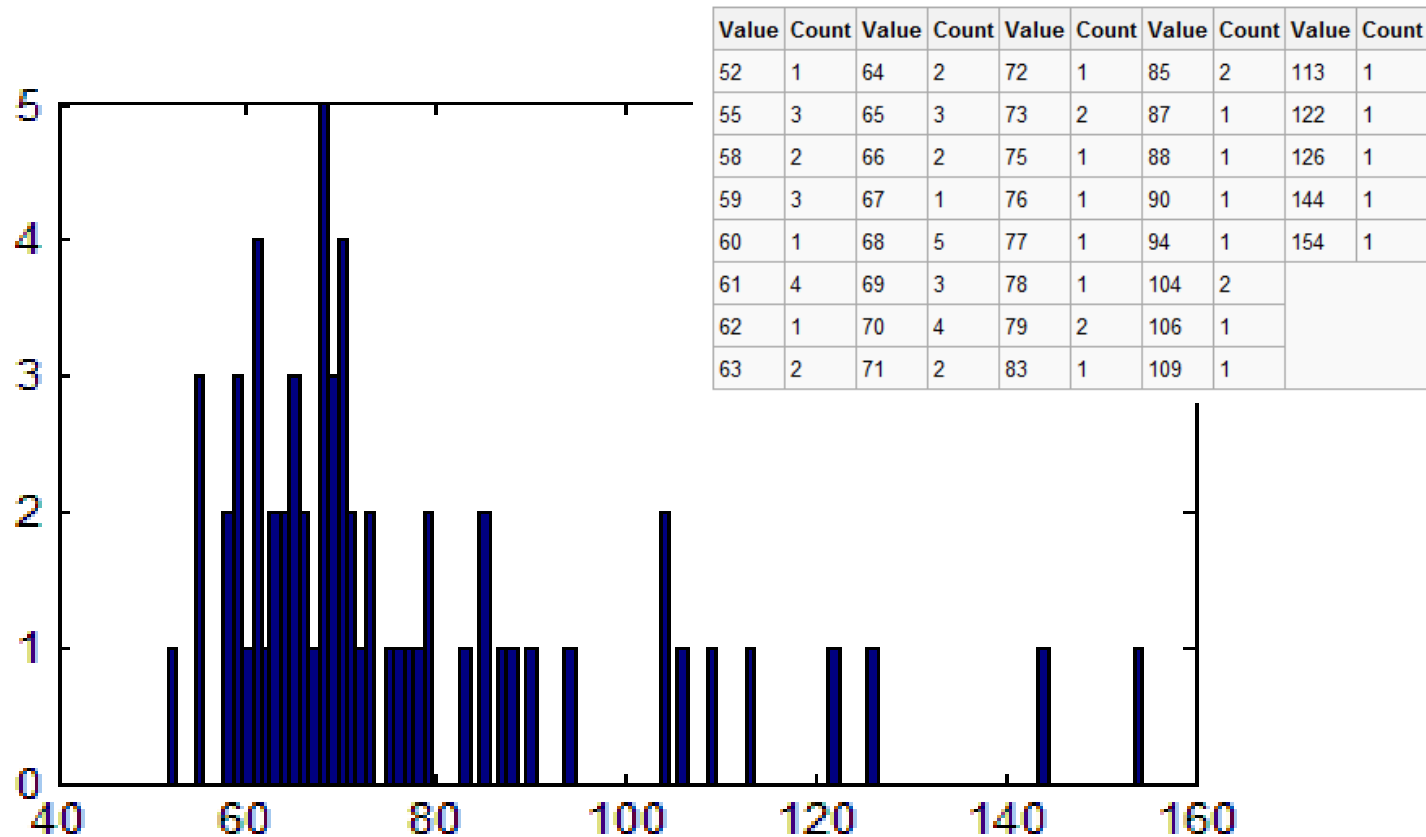
Image Histogram (Non-zero values)

Histogram Equalization: Example

Image Histogram (Non-zero values shown)

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Histogram Equalization: Example



Histogram Equalization: Example

Cumulative Distribution Function (cdf)

Image Histogram/Prob Mass Function

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Value	cdf	Value	cdf	Value	cdf	Value	cdf	Value	cdf
52		64		72		85		113	
55		65		73		87		122	
58		66		75		88		126	
59		67		76		90		144	
60		68		77		94		154	
61		69		78		104			
62		70		79		106			
63		71		83		109			

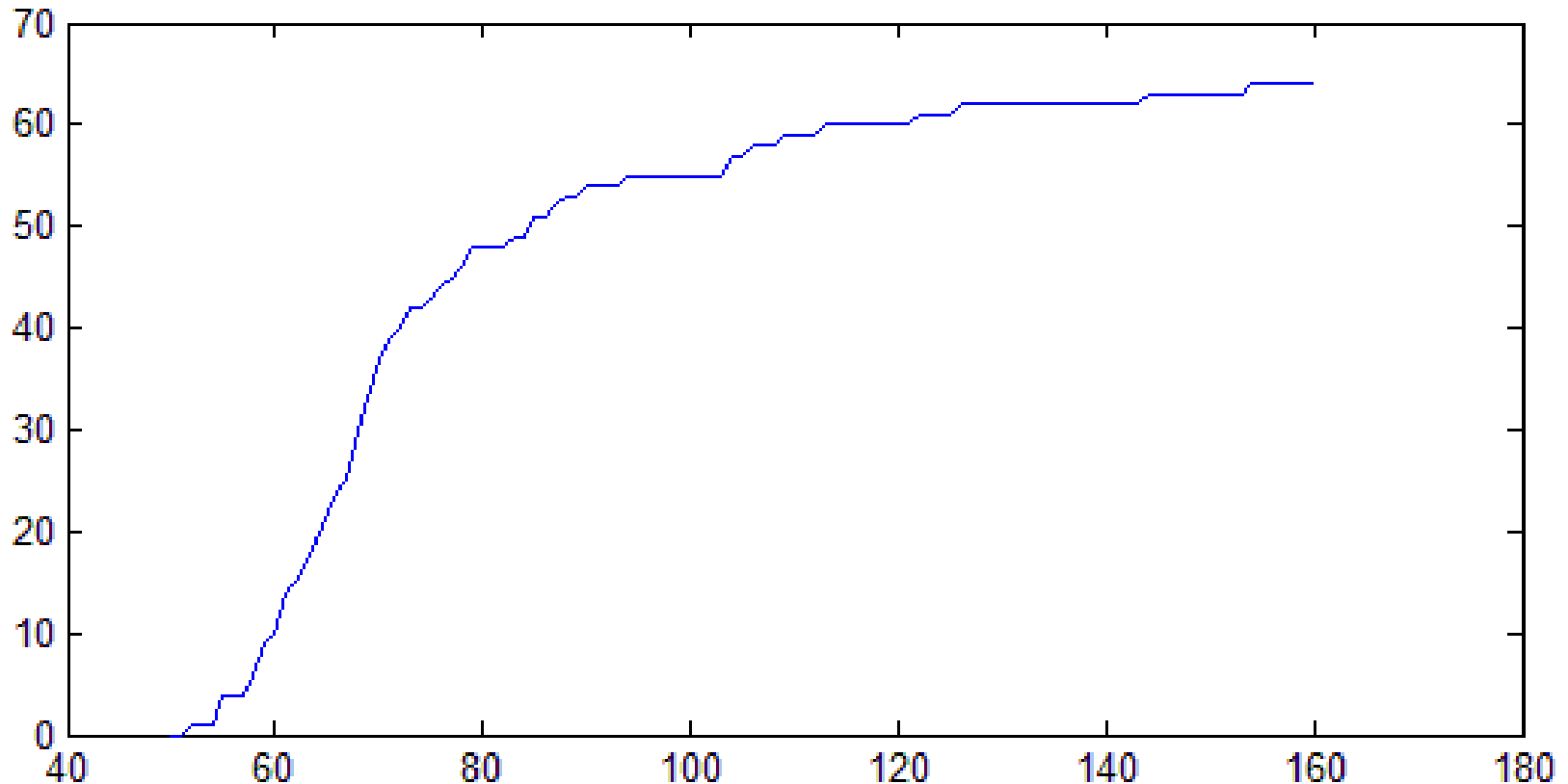
Histogram Equalization: Example

Cumulative Distribution Function (cdf)

Value	cdf	Value	cdf	Value	cdf	Value	cdf	Value	cdf
52	1	64	19	72	40	85	51	113	60
55	4	65	22	73	42	87	52	122	61
58	6	66	24	75	43	88	53	126	62
59	9	67	25	76	44	90	54	144	63
60	10	68	30	77	45	94	55	154	64
61	14	69	33	78	46	104	57		
62	15	70	37	79	48	106	58		
63	17	71	39	83	49	109	59		

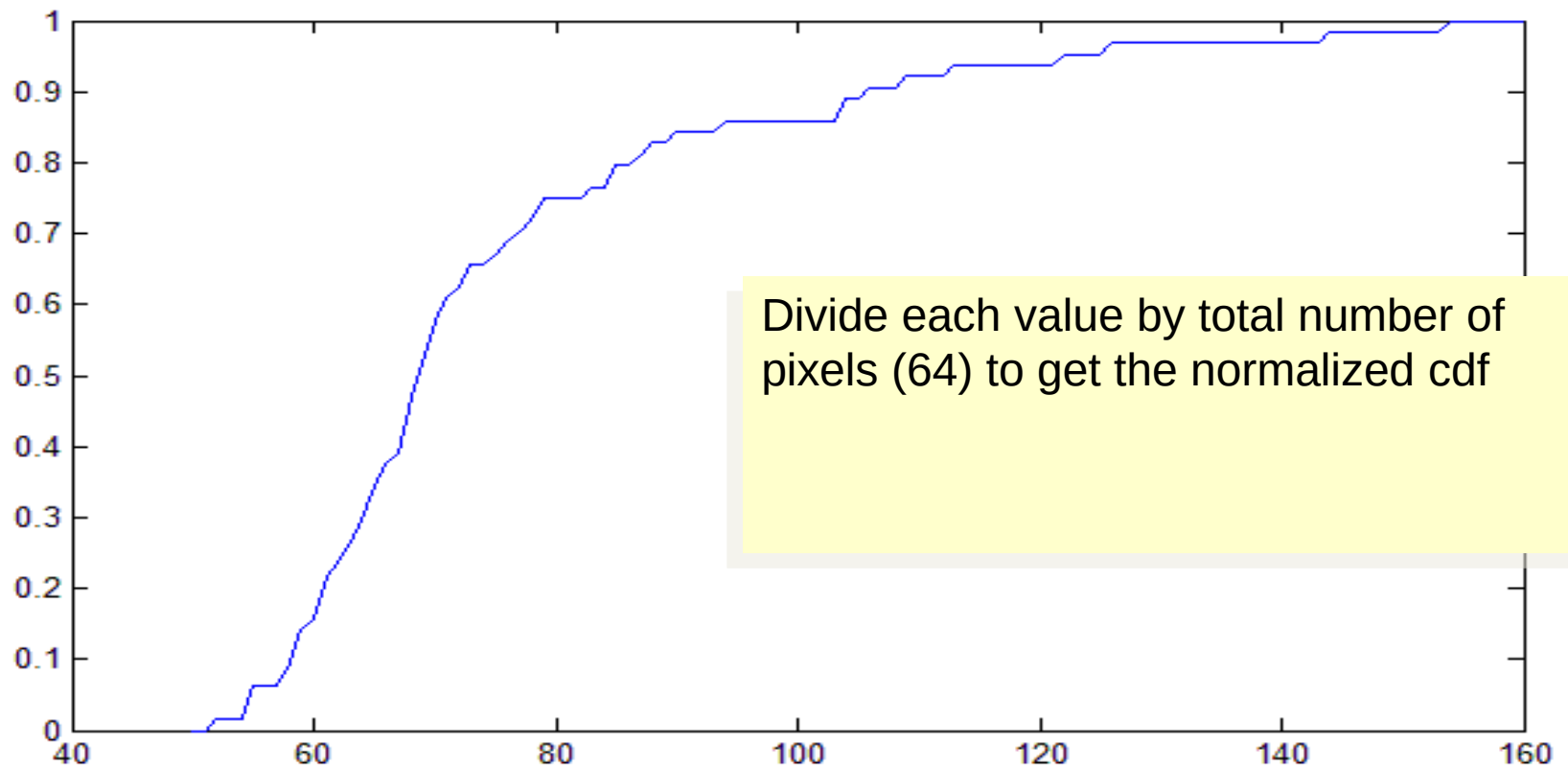
Histogram Equalization: Example

Cumulative Distribution Function (cdf)



Histogram Equalization: Example

Normalized Cumulative Distribution Function (cdf)



Histogram Equalization: Example

Value	cdf	Value	cdf	Value	cdf	Value	cdf	Value	cdf
52	1	64	19	72	40	85	51	113	60
55	4	65	22	73	42	87	52	122	61
58	6	66	24	75	43	88	53	126	62
59	9	67	25	76	44	90	54	144	63
60	10	68	30	77	45	94	55	154	64
61	14	69	33	78	46	104	57		
62	15	70	37	79	48	106	58		
63	17	71	39	83	49	109	59		

$$J(r, c) = 255 \cdot P_I[I(r, c)].$$

If cdf is normalized

$$s = \text{round}(255 \cdot \text{cdf}(r))$$

If cdf is NOT normalized

$$s = \text{round}(255 \cdot \frac{\text{cdf}(r)}{M \times N})$$

$$s = \text{round}(255 \cdot (46 / 64))$$

$$s = 183$$

52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

$M \times$

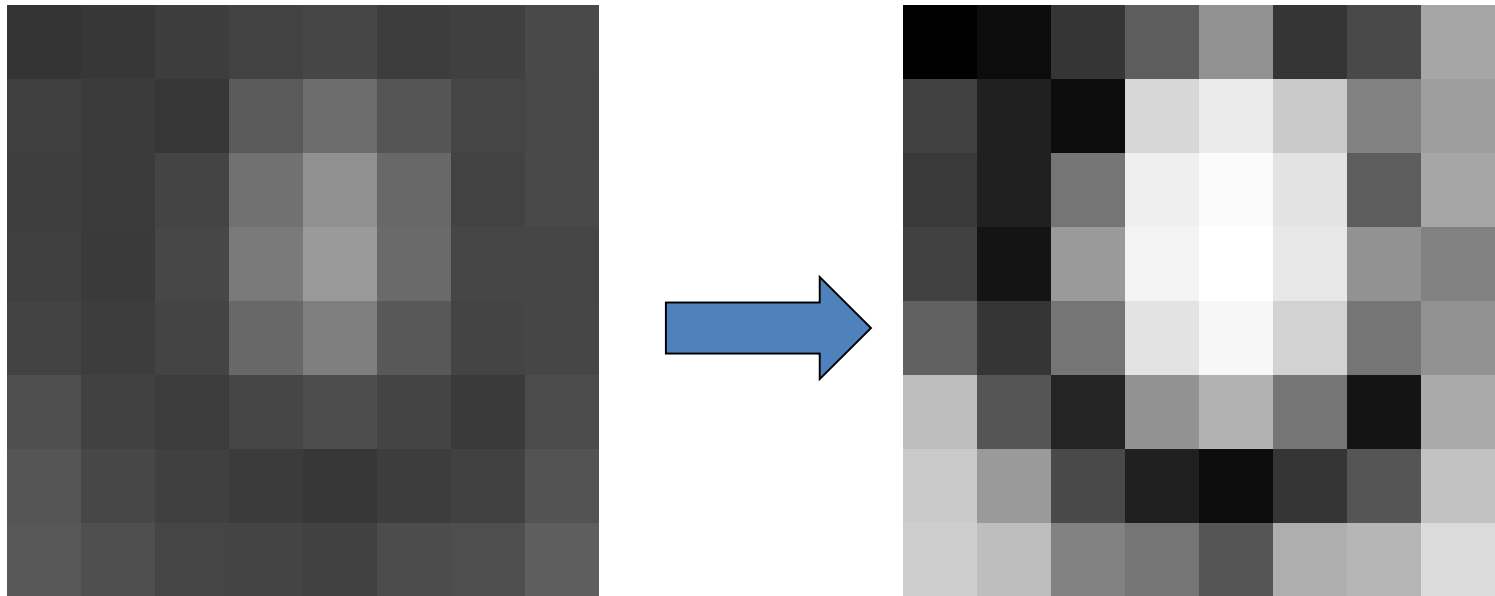
$s = \text{round}(255.(46 /$

$s = 183$

183

Original Image

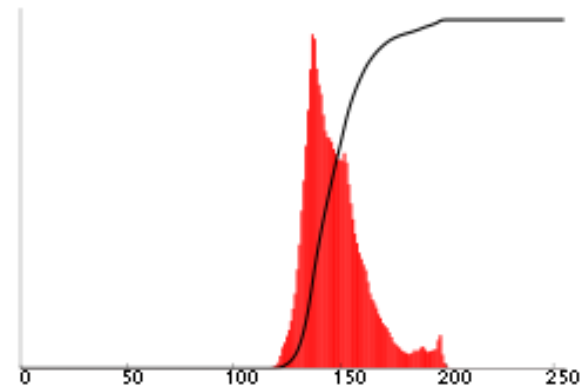
Histogram Equalization: Example



Histogram Equalization: Example



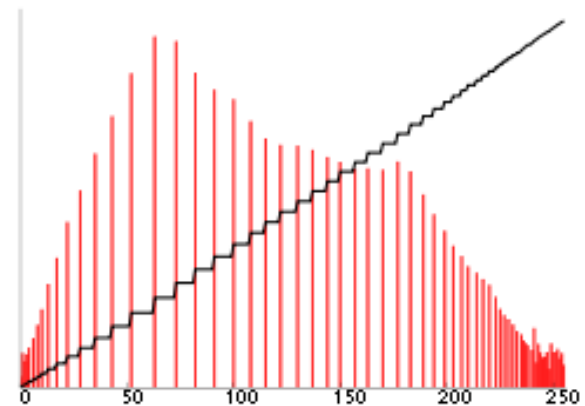
Original Image



Corresponding histogram (red) and cumulative histogram (black)



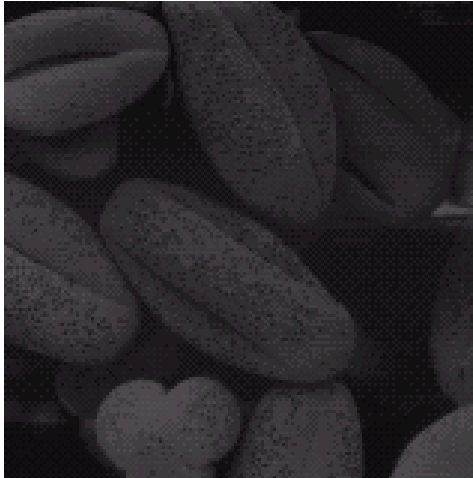
Image after histogram equalization



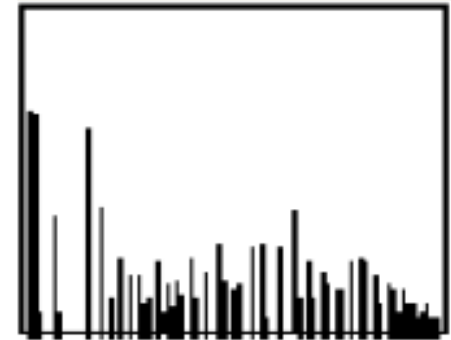
Corresponding histogram (red) and cumulative histogram (black)

Histogram Equalization: Example

Dark image



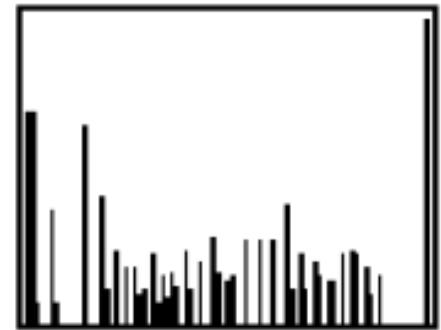
Equalized Histogram



Bright image



Equalized Histogram

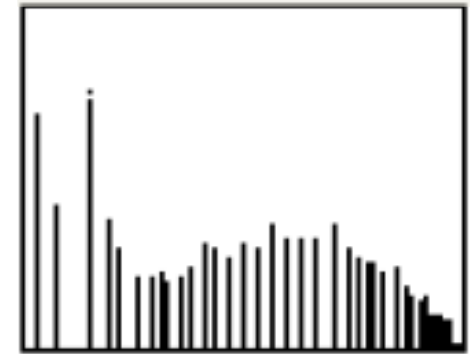


Histogram Equalization: Example

Low contrast



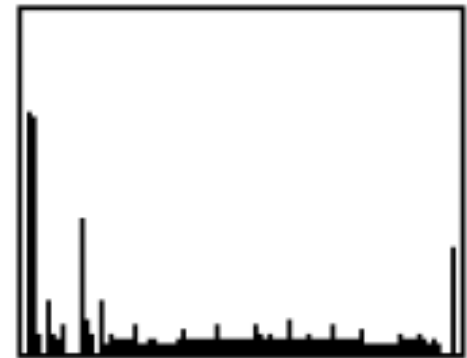
Equalized Histogram



High Contrast



Equalized Histogram



HISTOGRAM MATCHING (SPECIFICATION)

- HISTOGRAM EQUALIZATION DOES NOT ALLOW INTERACTIVE IMAGE ENHANCEMENT AND GENERATES ONLY ONE RESULT: AN APPROXIMATION TO A UNIFORM HISTOGRAM.
- SOMETIMES THOUGH, WE NEED TO BE ABLE TO SPECIFY PARTICULAR HISTOGRAM SHAPES CAPABLE OF HIGHLIGHTING CERTAIN GRAY-LEVEL RANGES.

HISTOGRAM SPECIFICATION

- THE PROCEDURE FOR HISTOGRAM-SPECIFICATION BASED ENHANCEMENT IS:
 - EQUALIZE THE LEVELS OF THE ORIGINAL IMAGE USING:

$$s = T(r_k) = \sum_{j=0}^k \frac{n_j}{n}$$

n: total number of pixels,

n_j: number of pixels with gray level r_j,

L: number of discrete gray levels

HISTOGRAM SPECIFICATION

- SPECIFY THE DESIRED DENSITY FUNCTION AND OBTAIN THE TRANSFORMATION FUNCTION $G(z)$:

$$v_k = G(z_k) = \sum_{i=0}^k p_z(z_i) = s_k$$

p_z : specified desirable PDF for output

HISTOGRAM SPECIFICATION

- THE NEW, PROCESSED VERSION OF THE ORIGINAL IMAGE CONSISTS OF GRAY LEVELS CHARACTERIZED BY THE SPECIFIED DENSITY $p_z(z)$.

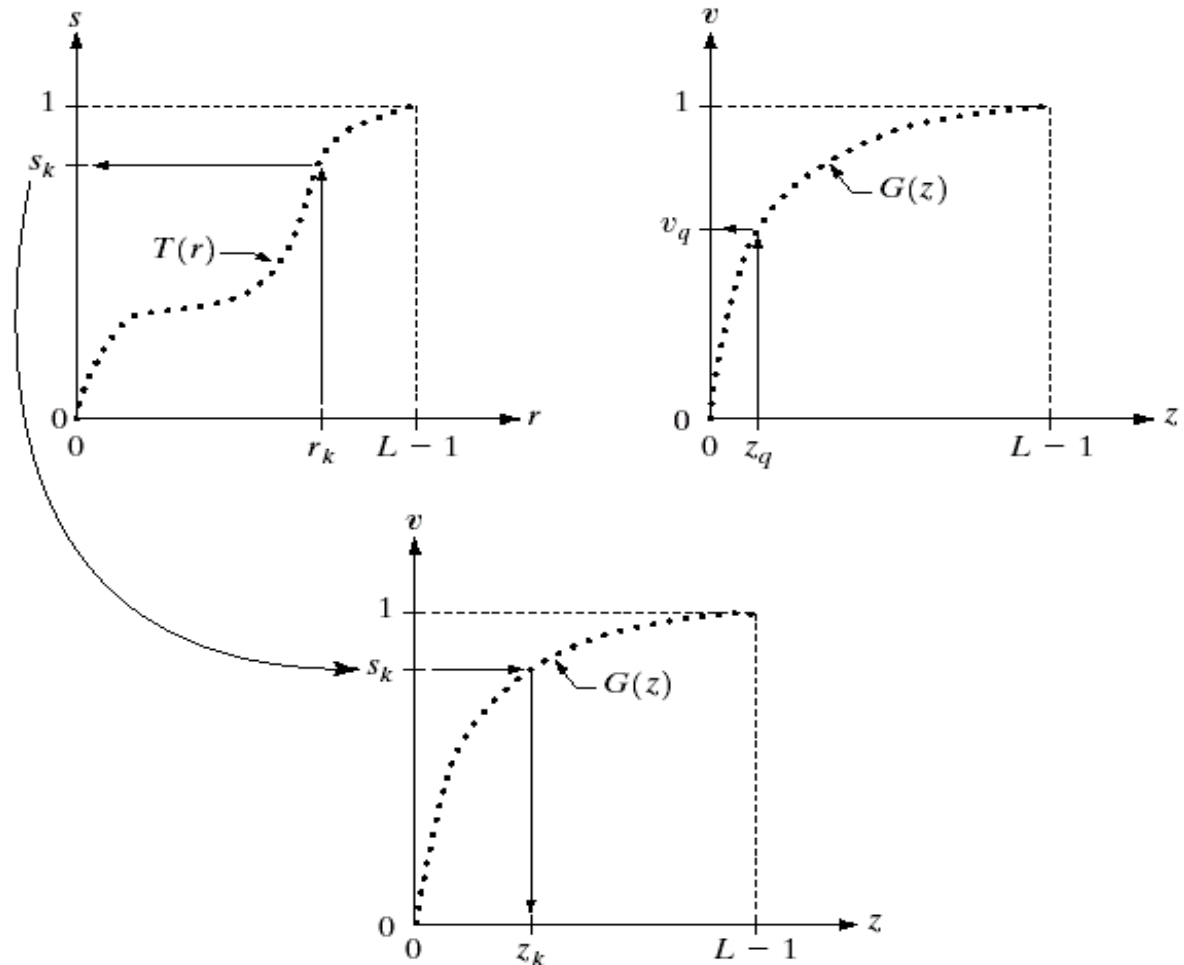
In essence: $z = G^{-1}(s) \rightarrow z = G^{-1}[T(r)]$

MAPPINGS

a b
c

FIGURE 3.19

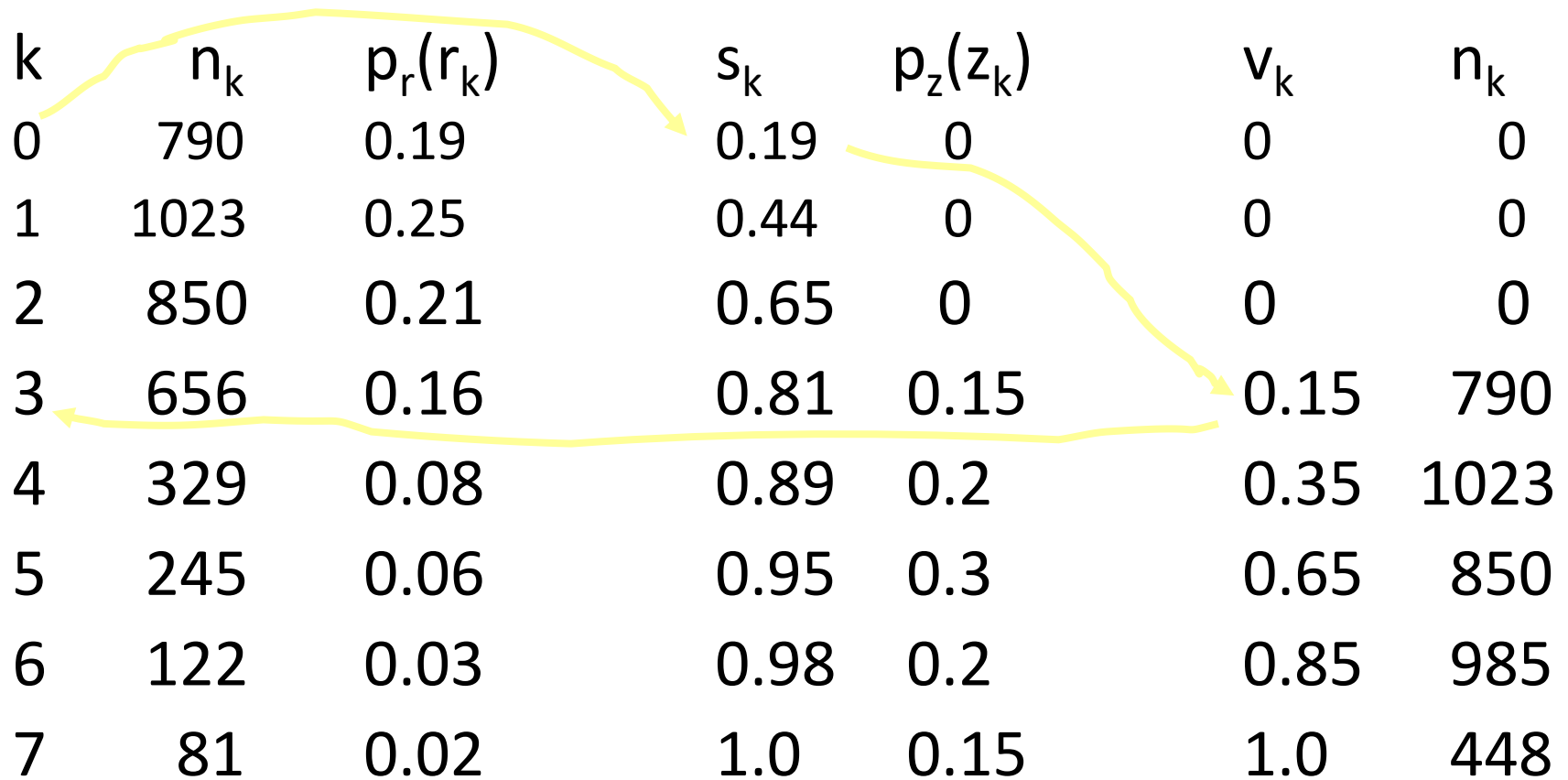
(a) Graphical interpretation of mapping from r_k to s_k via $T(r)$.
 (b) Mapping of z_q to its corresponding value v_q via $G(z)$.
 (c) Inverse mapping from s_k to its corresponding value of z_k .



HISTOGRAM SPECIFICATION

- OBTAIN THE HISTOGRAM OF THE GIVEN IMAGE
- MAP EACH LEVEL r_k TO A LEVEL S_k
- OBTAIN THE TRANSFORMATION FUNCTION G FROM THE GIVEN $P_z(z)$
- PRECOMPUTE z_k FOR EACH VALUE OF S_k
- FOR EACH PIXEL IN THE ORIGINAL IMAGE, IF THE VALUE OF THAT PIXEL IS r_k MAP THIS VALUE TO ITS CORRESPONDING LEVEL S_k , THEN MAP LEVEL S_k INTO THE FINAL VALUE z_k

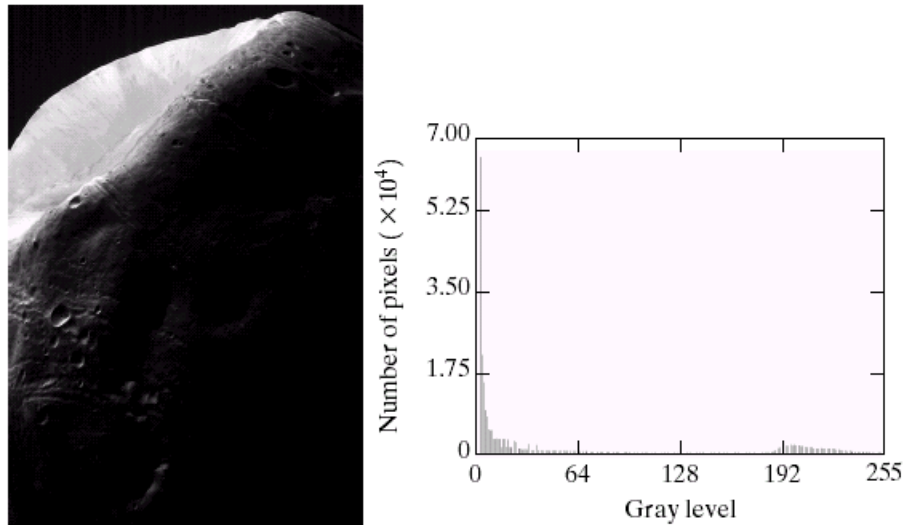
HISTOGRAM SPECIFICATION



k	n_k	$p_r(r_k)$	s_k	$p_z(z_k)$	v_k	n_k
0	790	0.19	0.19	0	0	0
1	1023	0.25	0.44	0	0	0
2	850	0.21	0.65	0	0	0
3	656	0.16	0.81	0.15	0.15	790
4	329	0.08	0.89	0.2	0.35	1023
5	245	0.06	0.95	0.3	0.65	850
6	122	0.03	0.98	0.2	0.85	985
7	81	0.02	1.0	0.15	1.0	448

A 64X64 (4096 PIXELS) IMAGE WITH 8 GRAY LEVELS

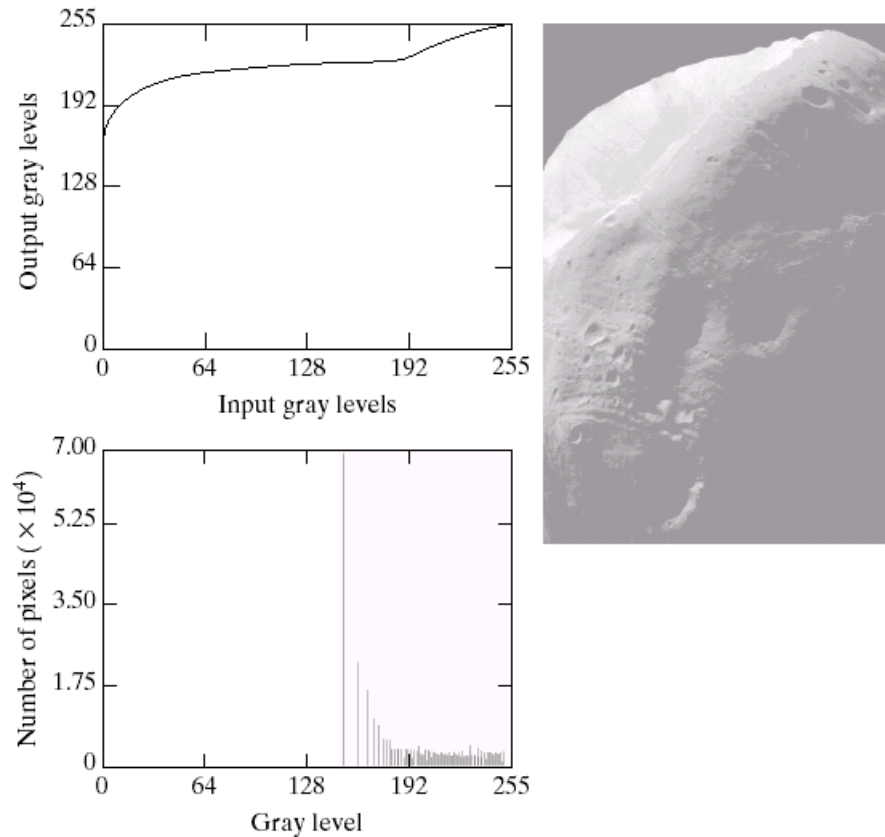
IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN



a b

FIGURE 3.20 (a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor*. (b) Histogram. (Original image courtesy of NASA.)

IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN



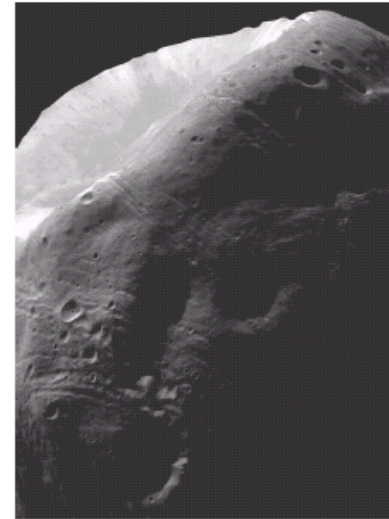
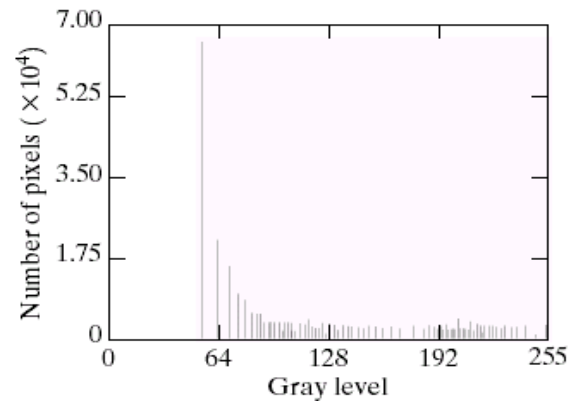
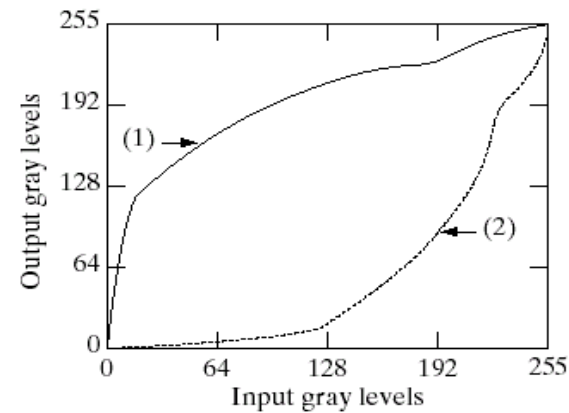
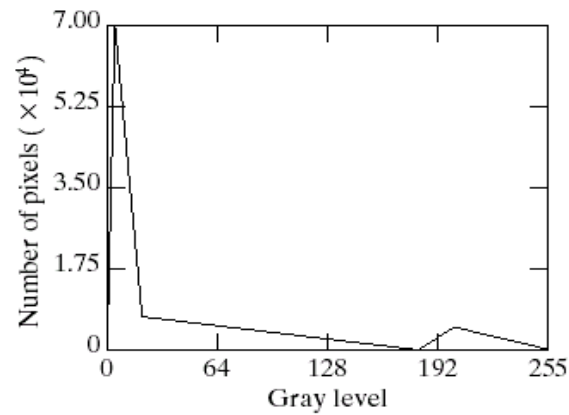
a b
c

FIGURE 3.21
(a) Transformation function for histogram equalization.
(b) Histogram-equalized image (note the washed-out appearance).
(c) Histogram of (b).

a c
b
d

FIGURE 3.22

(a) Specified histogram.
(b) Curve (1) is from Eq. (3.3-14), using the histogram in (a); curve (2) was obtained using the iterative procedure in Eq. (3.3-17).
(c) Enhanced image using mappings from curve (2).
(d) Histogram of (c).



GLOBAL/LOCAL HISTOGRAM EQUALIZATION

- IT MAY BE NECESSARY TO ENHANCE DETAILS OVER SMALL AREAS IN THE IMAGE
- THE NUMBER OF PIXELS IN THESE AREAS MAY HAVE NEGLIGIBLE INFLUENCE ON THE COMPUTATION OF A GLOBAL TRANSFORMATION WHOSE SHAPE DOES NOT NECESSARILY GUARANTEE THE DESIRED LOCAL ENHANCEMENT
- DEVISE TRANSFORMATION FUNCTIONS BASED ON THE GRAY LEVEL DISTRIBUTION IN THE NEIGHBORHOOD OF EVERY PIXEL IN THE IMAGE
- THE PROCEDURE IS:
 - DEFINE A SQUARE (OR RECTANGULAR) NEIGHBORHOOD AND MOVE THE CENTER OF THIS AREA FROM PIXEL TO PIXEL.
 - AT EACH LOCATION, THE HISTOGRAM OF THE POINTS IN THE NEIGHBORHOOD IS COMPUTED AND EITHER A HISTOGRAM EQUALIZATION OR HISTOGRAM SPECIFICATION TRANSFORMATION FUNCTION IS OBTAINED.
 - THIS FUNCTION IS FINALLY USED TO MAP THE GRAY LEVEL OF THE PIXEL CENTERED IN THE NEIGHBORHOOD.
 - THE CENTER IS THEN MOVED TO AN ADJACENT PIXEL LOCATION AND THE PROCEDURE IS REPEATED.

GLOBAL/LOCAL HISTOGRAM EQUALIZATION

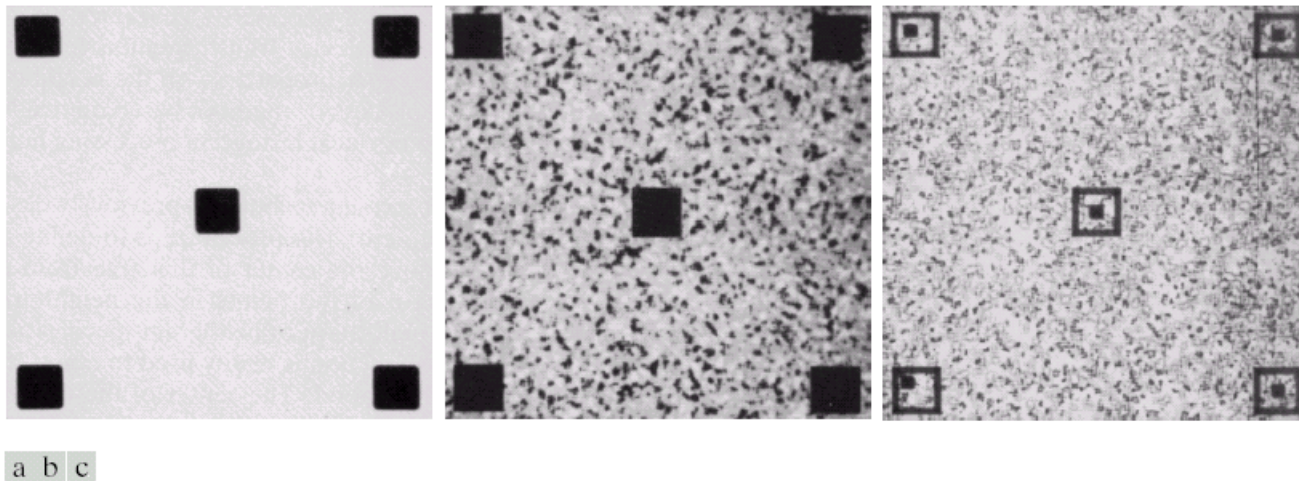


FIGURE 3.23 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization using a 7×7 neighborhood about each pixel.

USE OF HISTOGRAM STATISTICS FOR IMAGE ENHANCEMENT (Global)

- LET r REPRESENT A GRAY LEVEL IN THE IMAGE $[0, L-1]$, AND LET $p(r_i)$ DENOTE THE NORMALIZED HISTOGRAM COMPONENT CORRESPONDING TO THE i^{th} VALUE OF r .
- THE n^{th} MOMENT OF r ABOUT ITS MEAN IS DEFINED AS

$$\mu_n(r) = \sum_{i=0}^{L-1} (r_i - m)^n p(r_i)$$

- WHERE m IS THE MEAN VALUE OF r (AVERAGE GRAY LEVEL)

$$m = \sum_{i=0}^{L-1} r_i p(r_i)$$

USE OF HISTOGRAM STATISTICS FOR IMAGE ENHANCEMENT (Global)

- THE SECOND MOMENT IS GIVEN BY

$$\mu_2(r) = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

- WHICH IS THE VARIANCE OF r
- MEAN AS A MEASURE OF AVERAGE GRAY LEVEL IN THE IMAGE
- VARIANCE AS A MEASURE OF AVERAGE CONTRAST

USE OF HISTOGRAM STATISTICS FOR IMAGE ENHANCEMENT (Local)

- LET (x,y) BE THE COORDINATES OF A PIXEL IN AN IMAGE, AND LET $S_{x,y}$ DENOTE A NEIGHBORHOOD OF SPECIFIED SIZE, CENTERED AT (x,y)
- THE MEAN VALUE $m_{s_{xy}}$ OF THE PIXELS IN $S_{x,y}$ IS

$$m_{s_{xy}} = \sum_{(s,t) \in S_{xy}} r_{s,t} p(r_{s,t})$$

- THE GRAY LEVEL VARIANCE OF THE PIXELS IN REGION $S_{x,y}$ IS GIVEN BY

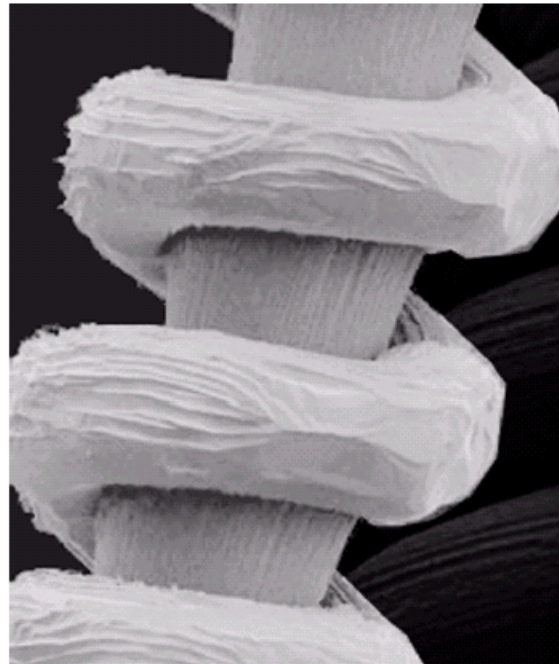
$$\sigma_{s_{xy}}^2 = \sum_{(s,t) \in S_{xy}} [r_{s,t} - m_{s_{xy}}]^2 p(r_{s,t})$$

USE OF HISTOGRAM STATISTICS FOR IMAGE ENHANCEMENT

- THE GLOBAL MEAN AND VARIANCE ARE MEASURED OVER AN ENTIRE IMAGE AND ARE USEFUL FOR GROSS ADJUSTMENTS OF OVERALL INTENSITY AND CONTRAST.
- A USE OF THESE MEASURES IN LOCAL ENHANCEMENT IS, WHERE THE LOCAL MEAN AND VARIANCE ARE USED AS THE BASIS FOR MAKING CHANGES THAT DEPEND ON IMAGE CHARACTERISTICS IN A PREDEFINED REGION ABOUT EACH PIXEL IN THE IMAGE.

TUNGSTEN FILAMENT IMAGE

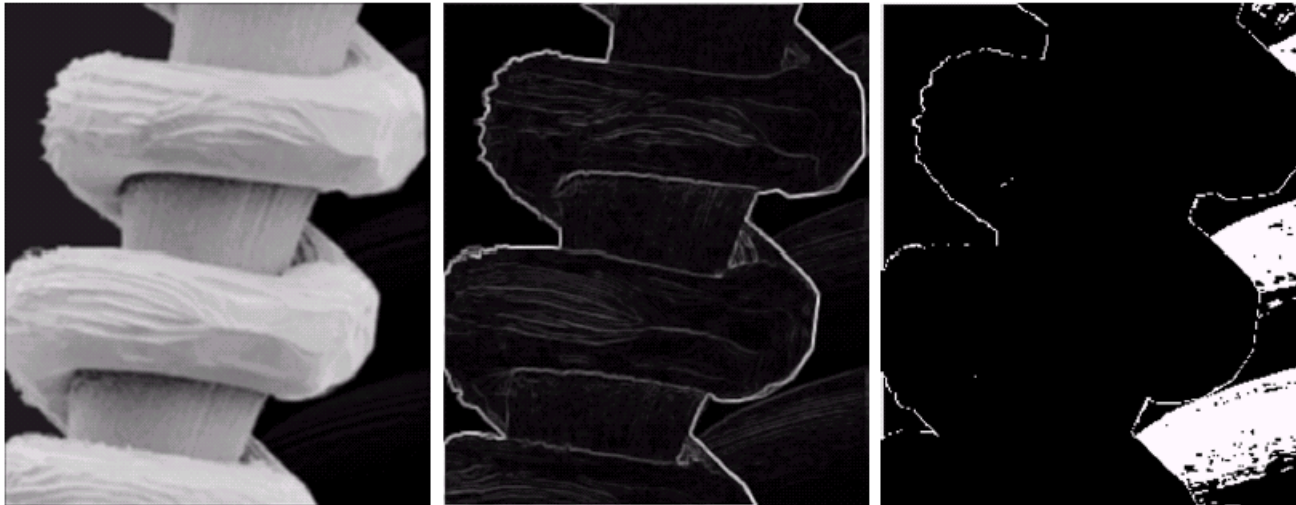
FIGURE 3.24 SEM image of a tungsten filament and support, magnified approximately 130 $\times$. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene).



USE OF HISTOGRAM STATISTICS FOR IMAGE ENHANCEMENT

- A PIXEL AT POINT (x,y) IS CONSIDERED IF:
 - $m_{sxy} \leq k_0 M_G$, where k_0 is a positive constant less than 1.0, and M_G is global mean
 - $\sigma_{sxy} \leq k_2 D_G$, where D_G is the global standard deviation and k_2 is a positive constant
 - $k_1 D_G \leq \sigma_{sxy}$, with $k_1 < k_2$
- A PIXEL THAT MEETS ALL ABOVE CONDITIONS IS PROCESSED SIMPLY BY MULTIPLYING IT BY A SPECIFIED CONSTANT, E, TO INCREASE OR DECREASE THE VALUE OF ITS GRAY LEVEL RELATIVE TO THE REST OF THE IMAGE.
- THE VALUES OF PIXELS THAT DO NOT MEET THE ENHANCEMENT CONDITIONS ARE LEFT UNCHANGED.

IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN



a b c

FIGURE 3.25 (a) Image formed from all local means obtained from Fig. 3.24 using Eq. (3.3-21). (b) Image formed from all local standard deviations obtained from Fig. 3.24 using Eq. (3.3-22). (c) Image formed from all multiplication constants used to produce the enhanced image shown in Fig. 3.26.

IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN



FIGURE 3.26
Enhanced SEM
image. Compare
with Fig. 3.24. Note
in particular the
enhanced area on
the right side of
the image.

Readings from Book (3rd Edn.)

- 3.3 Histogram



Acknowledgements

- ◆ Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- ◆ Pattern Recognition and Analysis Course – A.K. Jain, MSU
- ◆ *Pattern Classification*” by Duda et al., John Wiley & Sons.
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- ◆ Machine Vision: Automated Visual Inspection and Robot Vision”, David Vernon, Prentice Hall, 1991
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